Monday 2-5

From Order to Chaos: Measuring Randomness Across Card Shuffling Methodologies

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2.671 Measurement and Instrumentation
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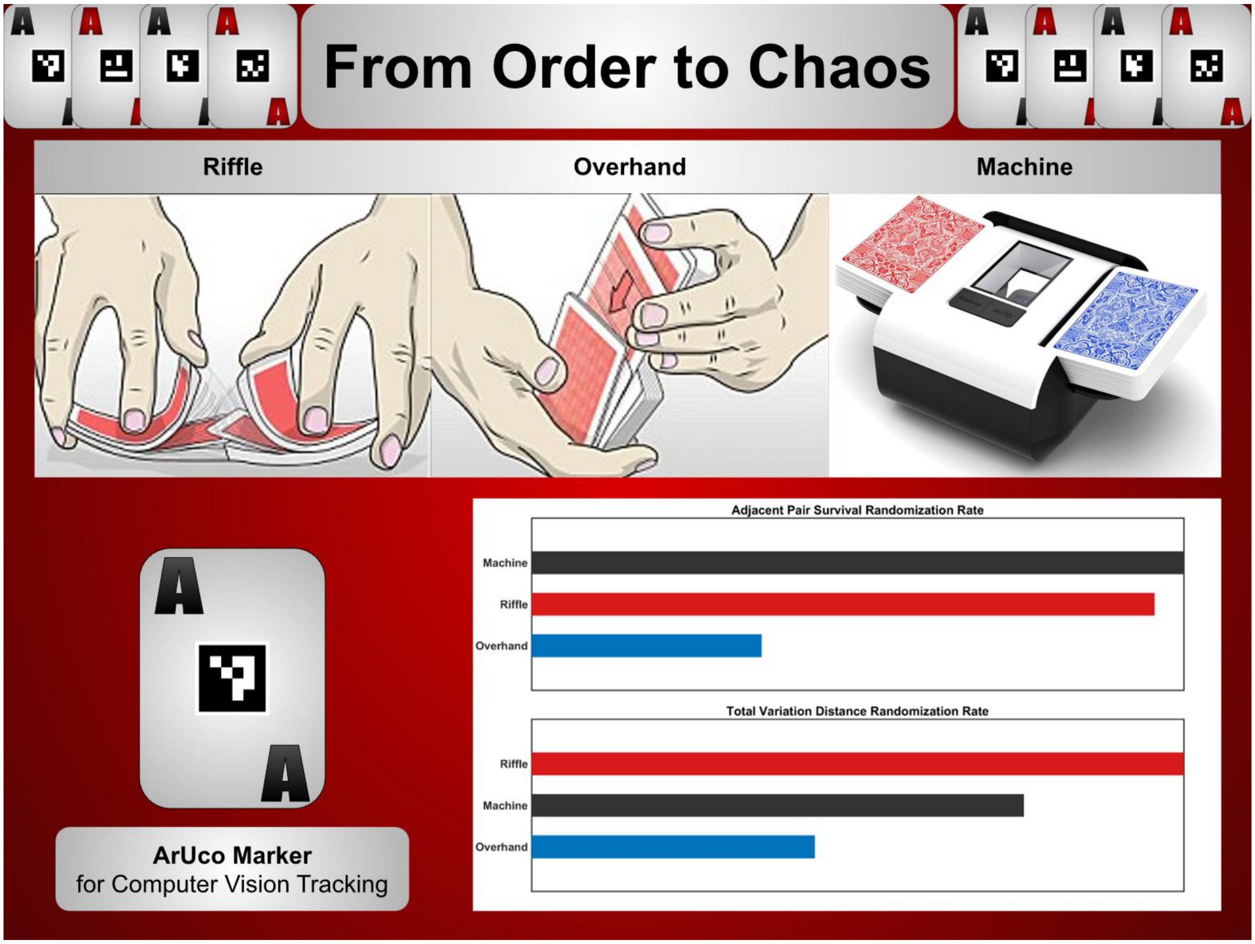
Visual Abstract



Abstract

Whether in casinos or at kitchen tables, countless card games rely on shuffling to ensure fair play, yet limited data exists on how different shuffle methods achieve true randomness. The effectiveness of riffle, overhand, and machine shuffling methods was quantified using statistical metrics derived from experimental trials by tracking how each card's position changed after every shuffle. A custom computer vision system using ArUco fiducial markers tracked all 52 card positions across 210 trials at shuffle counts $k = 0 - 7$. Randomness was evaluated through positional histograms, Adjacent-Pair Survival (APS), and normalized Total Variation (TV) distance. Riffle and machine shuffles approached the theoretical baseline of two surviving pairs with APS randomization rates of 1.49 and 1.56 per shuffle, respectively, while the overhand shuffle mixed more slowly at 0.55 per shuffle. Global TV analysis revealed decay rates of 0.53 for the riffle, 0.40 for the machine, and 0.23 for the overhand shuffle, indicating that full-deck randomness is achieved within roughly seven riffles but remains incomplete after seven overhands. These findings confirm theoretical predictions that riffle-style shuffles randomize far faster than overhand techniques, providing experimental evidence that links mathematical mixing theory to real-world shuffling behavior.

Keywords: Card Shuffling, Variation Distance, Randomization of Cards, Pair Adjacency, Fiducial Markers

1. Introduction

Every game that uses playing cards, from simple games like Uno to complex ones such as poker, relies on one simple expectation: that the deck is fair and unpredictable. The integrity of this randomness underpins both casual play and a global gambling industry worth nearly $288 billion [1]. To maintain fairness, casinos employ professional dealers, multiple decks, and automated shuffling machines to prevent any predictable patterns. Yet despite these efforts, it remains unclear which number or type of shuffle is needed to achieve a truly random deck. This study compares three common shuffling methods: the riffle shuffle, which interleaves two halves of the deck; the overhand shuffle, which transfers small packets of cards in a deck reversal; and an automatic machine shuffle, which mechanically interleaves cards in a consistent riffle-like pattern.

The mathematics of shuffling has been studied extensively, most famously in the work of Bayer and Diaconis [2], who showed that about seven riffle shuffles are theoretically sufficient to randomize a deck, and by Pemantle [3], who theorized that more than 1,000 overhand shuffles are necessary to reach a uniform distribution. However, most of the literature is theoretical, with comparatively little experimental investigation of how different shuffling methods perform in practice.

This project seeks to quantify the relationship between shuffle method, number of shuffles, and the resulting level of randomness in a standard deck of cards, to offer practical insight into how to best achieve fair play through shuffling. Three shuffle types are examined: riffle shuffle, overhand shuffle, and automatic machine shuffle. For each method, a standard 52-card deck is shuffled between one and seven times, and the resulting order is recorded after each shuffle. Each card is labeled with a unique ArUco visual-referent marker, similar to a QR code, and photographed in a fixed overhead setup to enable automated, precise tracking assisted by a custom computer-vision program. The full procedure is repeated ten times per method, producing 210 datasets for analysis. Randomness is quantified using positional histograms, which show how card positions change relative to their original order after each shuffle; mean variation in position across all cards; and adjacent-pair survival rates, which represent the fraction of original neighbors that remain together. Results are summarized across trials to estimate means and confidence intervals and are compared against theoretical baselines such as the expected convergence to randomness after seven riffle shuffles [2] as well as the 2/N adjacent-pair survival level predicted for a fully randomized deck [4]. This analysis provides quantitative evidence about the relative effectiveness of human and machine shuffle methods and demonstrates how experimental measurement can test theoretical predictions about randomness and fairness in card play. The project is expected to confirm that riffle shuffles are more effective and efficient at approaching a random distribution than overhand shuffling [2, 3].

2. Background

This section summarizes the different physical shuffle types examined in the experiment and provides the context needed to understand how their mixing behaviors can be evaluated. The mathematics of modeling a shuffle are defined along with the metrics used in this project to quantify randomness.

2.1 Types of Shuffles

Card shuffling can be accomplished through several distinct physical methods that differ in both mechanism and the degree of randomness they produce. The approaches studied in this project are the riffle shuffle, overhand shuffle, and automatic machine shuffle. Each uses a distinct physical process to mix the cards. The following image, Figure 1, demonstrates the visual technique of the three shuffles used in this project.

[5,6]

Figure 1: Visual comparison of the three shuffle methods examined in this study. The images show a human-performed riffle shuffle, an overhand shuffle, and an automated machine shuffle, each representing a distinct physical mechanism for mixing a deck. The riffle shuffle interleaves two halves, the overhand shuffle transfers sequential packets, and the machine shuffle reproduces a consistent mechanical interleave. This illustrates the fundamental differences in how each method moves cards through the deck, which directly affects the rate at which positional order breaks down, and randomness develops in the results that follow.

The riffle shuffle, the most common and well-known technique, is performed by dividing the deck into two roughly equal halves and releasing small packets from each half so that the cards interleave alternately. Gilbert [7] formalized this process in the Gilbert–Shannon–Reeds (GSR) model, which treats the interleaving as a probabilistic event. This model forms the basis for most mathematical analyses of card mixing and underlies the theoretical result by Bayer and Diaconis [2] that approximately seven riffle shuffles are sufficient to achieve near-random order for a 52-card deck. More generally, the number of shuffles needed for sufficient mixing scales approximately as $\frac{3}{2}log_2(N)$, where N is the number of cards in the deck. Subsequent work has extended this theoretical framework, including analyses of decks with repeated cards [8] and studies of the sharp transition, or cutoff, between under-shuffled and well-shuffled states [9]. These extensions illustrate how sensitive mixing behavior can be to deck, but they focus on simulated rather than physical shuffles. The lack of real physical shuffles motivates this project's study of how different physical shuffles compare to theoretical predictions.

The overhand shuffle transfers small groups of cards from one hand to another, usually by repeatedly sliding packets from the top of the deck to the receiving hand. Because packets remain largely intact, this method primarily produces local rearrangements, making it slower to randomize the deck. Pemantle [3] modeled the overhand shuffle and found that the expected number of shuffles required to achieve a uniform distribution scales approximately as $N^2\log(N)$. For a 52-card deck, this corresponds to more than a thousand shuffles, illustrating its relative inefficiency compared with the riffle shuffle's logarithmic rate of convergence.

An automatic machine shuffle uses mechanical components to displace cards from below by rotating wheels to split and interleave cards in an automated riffle-like process. These machines are designed to improve consistency and eliminate human bias, yet their internal mechanics may still introduce subtle, predictable patterns or non-uniform interleaving [10]. Measuring the randomness of their output provides insight into how well a mechanical system can emulate human behavior and the balance between random chance and mechanical precision.

Together, these shuffle types span a range from human to mechanical control, allowing direct experimental comparison of how physical implementation influences mixing. Previous experimental studies have focused mainly on riffle shuffles. Silverman [10] tracked the order of decks subjected to repeated riffle shuffles and analyzed permutation statistics such as rising and descending sequences to evaluate mixing behavior. His work confirmed progressive randomization but also showed that human variability leads to differences from idealized theoretical models. However, overhand shuffles and mechanical shufflers remain largely unexamined in physical experiments. This project addresses that gap by applying computer vision as a measurement tool rather than as a modeling approach and by comparing all three shuffle types within a unified measurement framework.

2.2 Modeling a Random Shuffle

Understanding how these physical shuffles mix a deck requires a measurement approach that captures the complete permutation of cards after each shuffle. This project uses computer vision with ArUco fiducial markers to automatically detect and record the position of every card, enabling a high-throughput, fully physical experiment without relying on simulation or manual transcription.

A card shuffle can be described mathematically as a permutation of the 52 cards in a deck. Because there are $52!$ or about 8×10^{67} possible orders to measure how close a given shuffle is to uniform randomness, randomness can instead be determined by analyzing how close a real shuffle is to a uniform random distribution of all possible orders [3]. Bayer and Diaconis used the total variation distance, defined in Eq. (1), between the shuffle distribution Q and the uniform distribution U:

$$Total\ Variation\ Distance: TV = \frac{1}{2}\sum_{i} |Q(i) - U(i)| \qquad (1)$$

This metric ranges from 0 (perfectly random) to 1 (completely ordered). Their analysis revealed a sharp cutoff near seven riffle shuffles, which has become the standard theoretical benchmark for "sufficient mixing" [2]. The theoretical cutoff described by Bayer and Diaconis provides a direct comparison point for this project, to test whether experimental data collected from physical shuffles follow the same convergence behavior. This project normalized Eq. (1) to find total variation distance as one of the quantifications of randomness.

While total variation distance is mathematically rigorous, it cannot be computed easily from experimental data due to the enormous number of permutations required to construct a full probability distribution. To evaluate real shuffled decks, researchers therefore rely on observable metrics that capture how much order remains after each shuffle. In this project, randomness is quantified by using three complementary measures. Positional histograms track how far each card moves from its starting position, showing how the distribution of card locations spreads toward randomness and revealing how spatial structure breaks down as shuffling progresses. Adjacent-pair survival (APS) measures the fraction of neighboring cards that remain next to each other after shuffling, providing a direct assessment of how quickly local order is destroyed. For a perfectly random 52-card deck, this value approaches $2/N \approx 0.038$ [2, 4]. Normalized Total Variation (TV) distance evaluates global randomness by comparing the measured per-card positional distributions to a uniform random model, providing a quantitative measure of how closely the deck approaches true statistical randomness.

These three metrics were selected because together they capture visual spread, local mixing, and global randomization in the deck, offering a comprehensive picture of how structure decays under repeated shuffling. These metrics can also be directly computed from the procedures used in this project: each card's location is extracted from photographs using ArUco markers, enabling full-deck statistical comparison after each shuffle.

3. Experimental Design

The experimental design consists of three components: the workflow used to collect shuffled deck data, the imaging setup that enabled consistent detection of all 52 cards, and the computer-vision pipeline that reconstructed each deck permutation. Together, these procedures generated a complete, automated record of card positions across hundreds of shuffles, providing the dataset used to quantify randomness in the results that follow.

3.1 Experimental Flow

Card-order randomness was quantified using a vision-based measurement system designed to track the positions of all 52 cards after repeated shuffles. The deck was arranged in a fixed 13×4 layout and imaged after each shuffle count in order to extract the full permutation of card identities. Data were collected for shuffle counts $k = 0$–7 and repeated across ten independent trials for each shuffling method to characterize trial-to-trial variability. The broad setup can be seen in Figure 2 shown below.

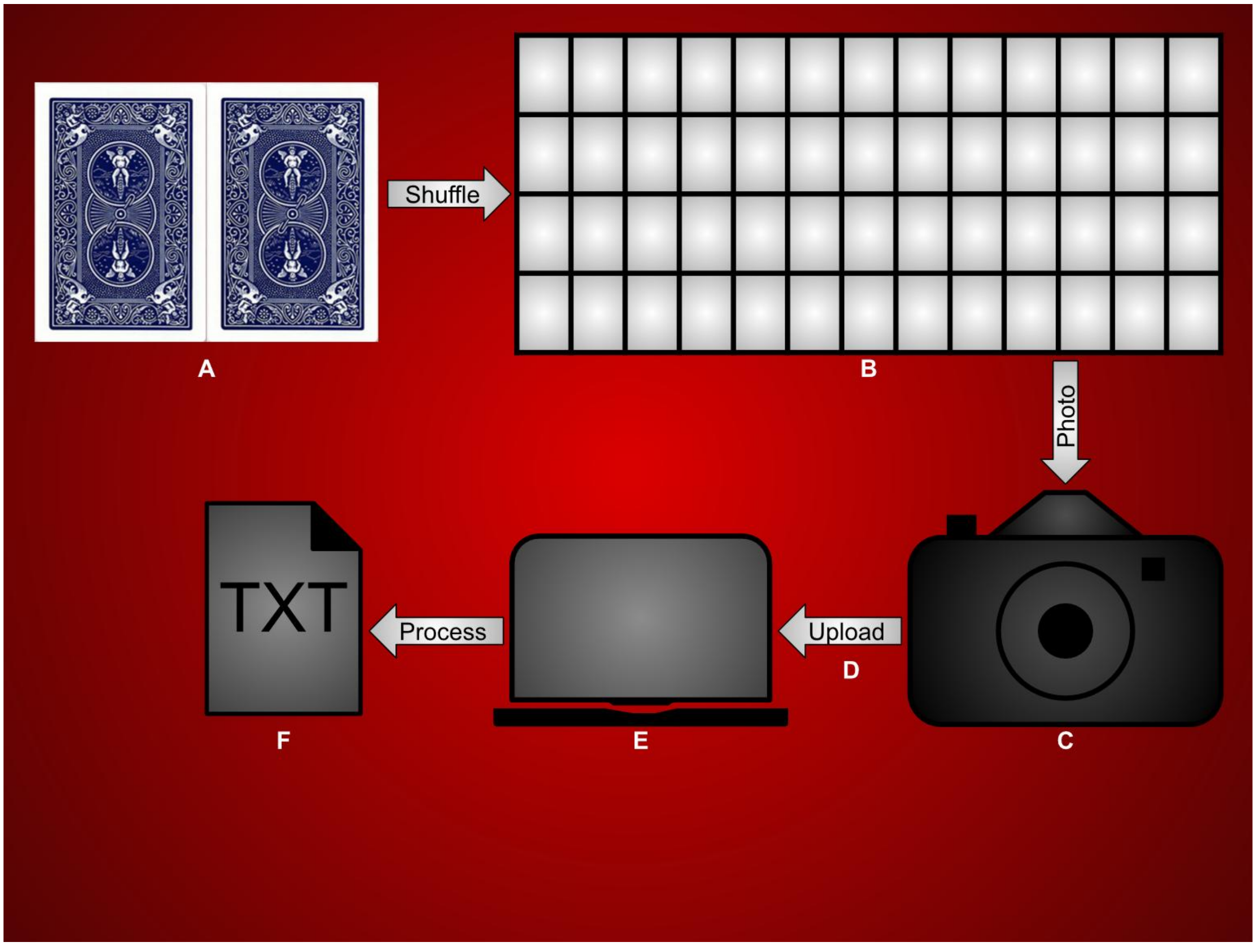


Figure 2: Experimental workflow: (A) Shuffle the deck (riffle/overhand/machine). (B) Lay cards face-up in a 13×4 grid. (C) Photograph from an overhead position. (D) Upload images to a laptop. (E) Run a Python script to detect card IDs and reconstruct the permutation. (F) Export a .txt file with the recovered order for analysis and plotting. This workflow enables fully automated, trial-to-trial-consistent measurement of every card's position, allowing high-throughput evaluation of how quickly different shuffle types destroy order and approach randomness.

3.2 Card Identification and Imaging Setup

Manual identification of card locations after every shuffle would be a time-intensive and error-prone process, especially across hundreds of trials, so an automated approach was required to reliably extract full deck permutations. Each playing card was fitted with a unique fiducial marker to allow automated identification. The markers were generated using an ArUco 4×4 dictionary [11] and printed as 1-inch by 1-inch squares on matte paper to minimize glare. Each marker was taped and centered on the face of its corresponding card to ensure full visibility in every captured image. Marker IDs 0–51 corresponded to card ranks Ace through King, ordered by suit as spades, hearts, clubs, and diamonds. An example of an ArUco marker attached to its corresponding card can be seen in Figure 3 below.

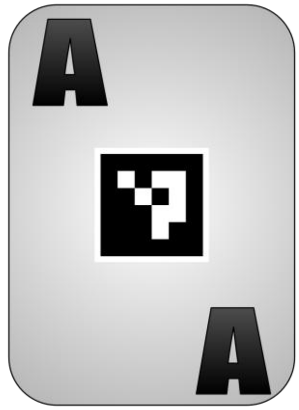


Figure 3: Example of an ArUco fiducial marker attached to a playing card. The image shows a single card fitted with its unique 4×4 ArUco marker, centered on the card face to ensure full visibility in overhead photographs. The black-and-white square pattern encodes the card's ID number, which the computer-vision script detects and uses to identify the card. This marker system is essential for the experiment because it enables reliable, automatic recognition of every card in every image, allowing full-deck permutations to be reconstructed consistently across all trials without manual labeling.

Following each shuffle, the deck was arranged face-up in thirteen columns and four rows on a matte white background. The cards were evenly spaced to prevent overlap and maintain consistent geometry across trials. Images were captured using a phone camera and subsequently uploaded to the custom computer-vision script for processing.

3.3 Computer Vision Analysis

Card identities and positions were extracted using a Python-based computer-vision script developed with OpenCV (version 4.3.0). Each image was processed by detecting the ArUco markers, decoding their unique IDs, and sorting them left-to-right and top-to-bottom to reconstruct the deck order. The resulting 52-element sequence was exported as a plain-text file containing the shuffle method, shuffle count, and trial number.

Images with fewer than 52 detections were re-captured to ensure data completeness. The resulting text files served as inputs for the quantitative randomness metrics outlined in the Background section, enabling computation of positional histograms, Adjacent-Pair Survival (APS), and normalized Total Variation (TV) distance.

4. Results and Discussion

Different shuffling techniques are examined to see how they transition a deck from ordered to random arrangements and how the measured randomization rates compare with theoretical predictions from card-mixing models. Three levels of analysis are conducted to characterize the evolution of randomness: positional histograms (Level 1) reveal how spatial patterns dissipate as cards diffuse across the deck; Adjacent-Pair Survival (APS) (Level 2) quantifies the decay of local order; and normalized Total Variation (TV) distance (Level 3) assesses the global convergence of the deck toward a uniform distribution. These progressively abstract metrics provide complementary perspectives on the transition from order to chaos and allow direct quantitative comparison of riffle, overhand, and mechanical shuffle dynamics to theoretical expectation.

Together, the results show that riffle and mechanical shuffles rapidly eliminate both local and global structure, while the overhand shuffle mixes far more slowly, preserving recognizable patterns even after multiple iterations. The following sections present these findings in detail and interpret how the physical mechanics of each shuffle method influence its effectiveness in producing randomness.

4.1 Positional Histograms

Positional histograms reveal the spatial evolution of card order as the shuffle count increases. At $k = 0$, the histograms show strong vertical bands corresponding to the perfectly ordered initial sequence. After a single riffle or machine shuffle, these bands begin to blur, forming diagonal streaks that indicate partial interleaving. By $k = 4$, the distributions appear visually uniform, with no discernible pattern beyond statistical noise.

In contrast, the overhand shuffle maintained distinct clusters even after multiple repetitions, with cards tending to remain near their original positions. This persistence of pattern demonstrates the low positional diffusion associated with the overhand technique. The visual transition from ordered bands to a scatter qualitatively confirms that riffle and machine shuffles destroy positional correlation far more efficiently. Figure 4, Figure 5 and Figure 6 reveal visible data for the change in card position.

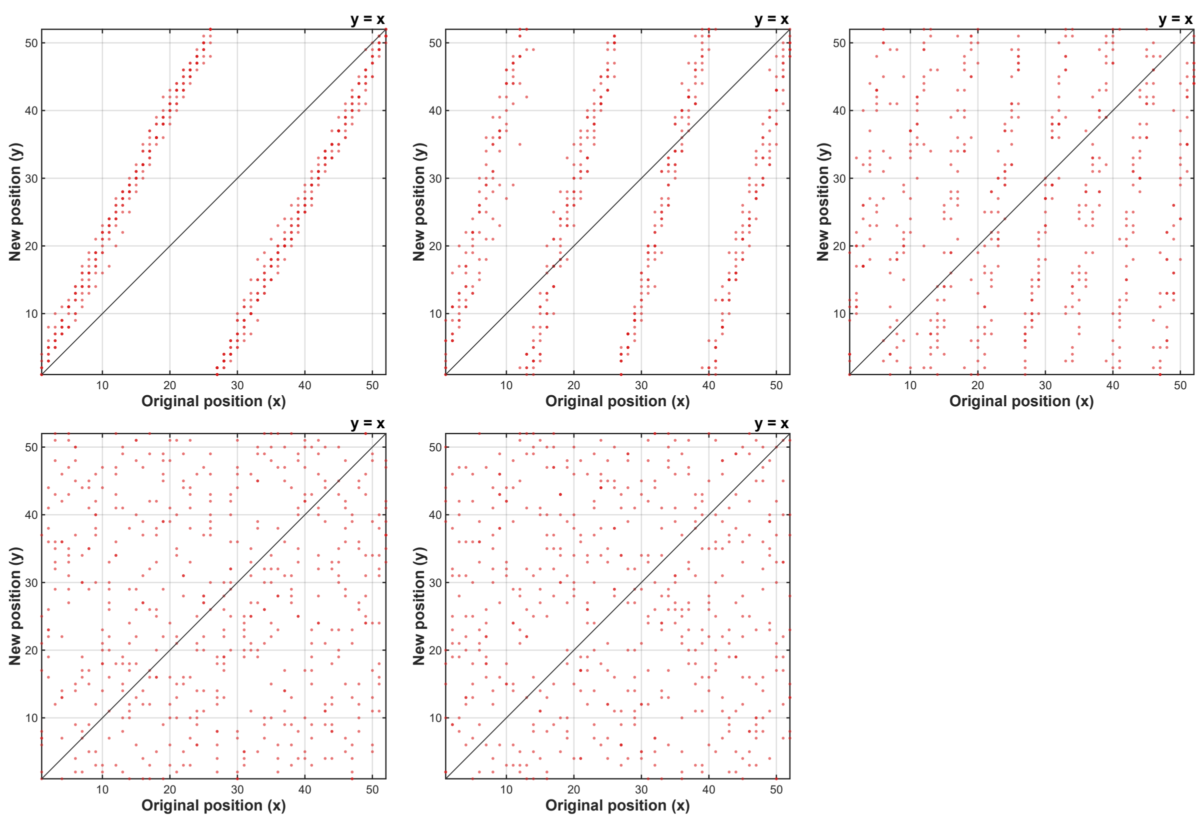


Figure 4: Positional Histograms for riffle shuffle $k = 1, 2, 3, 4, 7$. The riffle shuffle shows progressive dispersion of card positions as k increases. Patterns visible in the first three shuffles fade by k = 4, indicating substantial positional mixing. The line y = x provides a reference for cards that remained in the same location. All 10

trials are overlaid with a heat map indicating repeated position outcomes across trials.

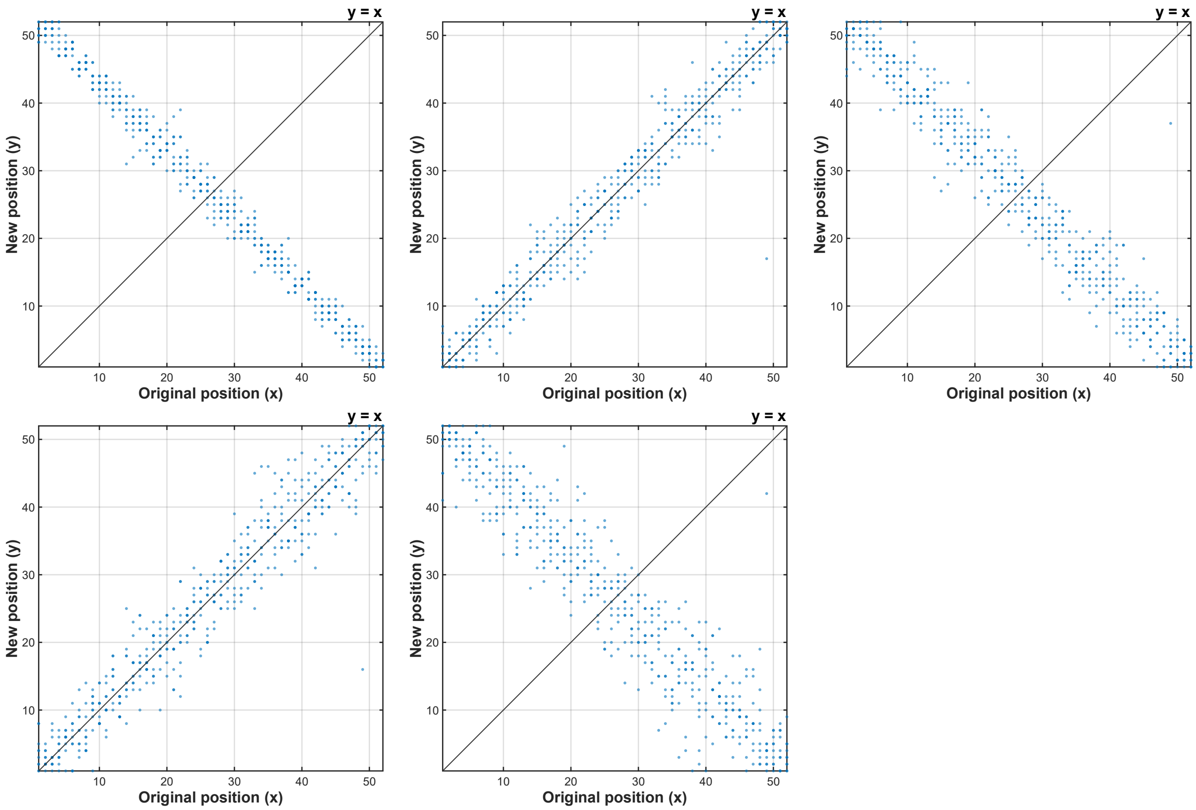


Figure 5: Positional Histograms for overhand shuffle $k = 1, 2, 3, 4, 7$. The overhand shuffle illustrates slow positional diffusion. Distinct structure remains visible at all shuffle counts, including the characteristic flip-flop pattern caused by partial reversal of packets during the shuffle. Although the bandwidth widens with repeated shuffling, qualitative randomness is not achieved. The line y = x indicates unchanged positions. All 10 trials are overlaid with a heat map showing repeated outcomes across trials.

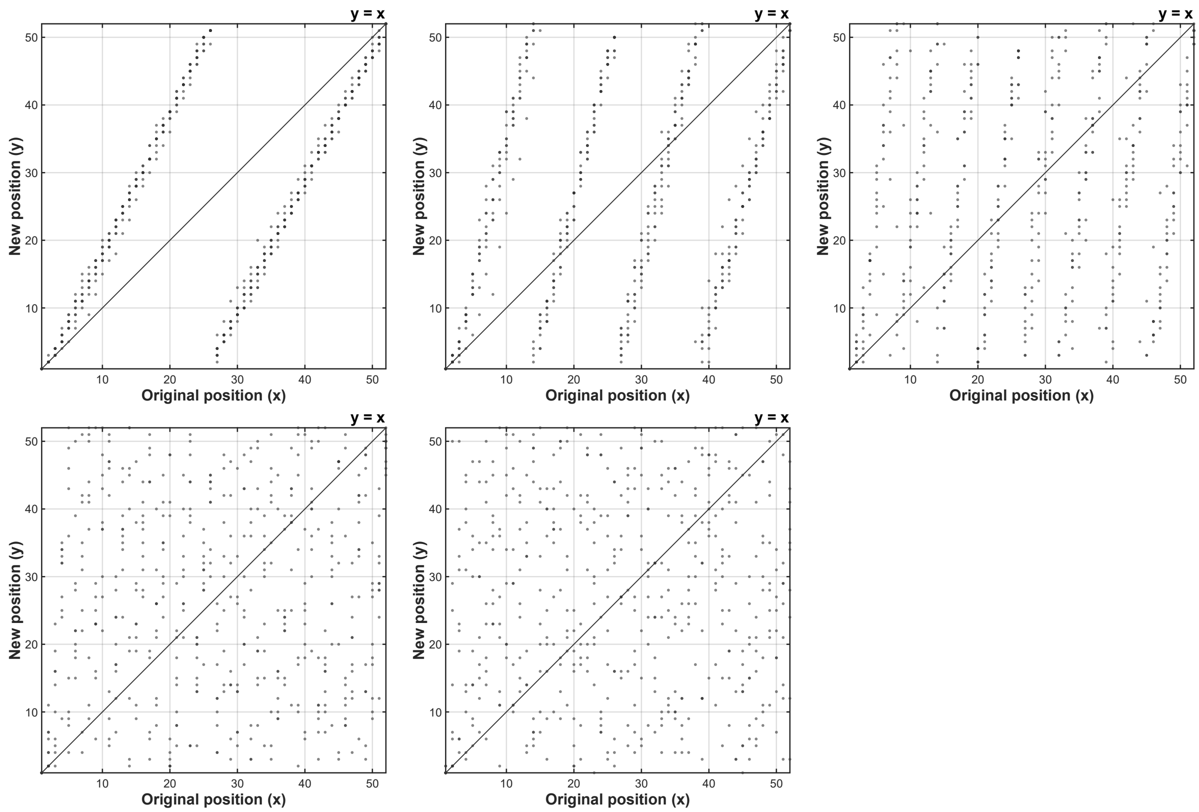


Figure 6: Positional Histograms for machine shuffle $k = 1, 2, 3, 4, 7$. Characterized by increasing visual qualitative randomness, distinct patterns present in the first 3 shuffles fade away. The line $y = x$ provides a reference for cards that remained in the same position. All 10 trials are plotted on this histogram with a heat map indicating when cards repeated positions across trials.

Taken together, the positional histograms show that riffle and machine shuffles rapidly disperse cards away from their original positions, eliminating recognizable structure within four shuffles. The overhand shuffle, however, exhibits slow and highly localized diffusion, with many cards remaining near their initial locations even after repeated mixing. This visual evidence establishes a key result of the study: riffle-style methods disrupt positional order far more efficiently than the overhand technique, setting the stage for the quantitative APS and TV analyses that follow.

4.2 Adjacent-Pair Survival (APS)

To quantify local order retention, the APS metric was computed as the number of initially neighboring card pairs that remained adjacent after k shuffles. The decay of adjacent pairs per shuffle was well-described by the exponential model seen in Eq. (2).

$$y = ae^{-bk} + c \tag{2}$$

where b represents the randomization rate per shuffle and c the theoretical random baseline. Figure 7 shows the Adjacent-Pair Survival rate across the different shuffling methodologies and shuffle counts.

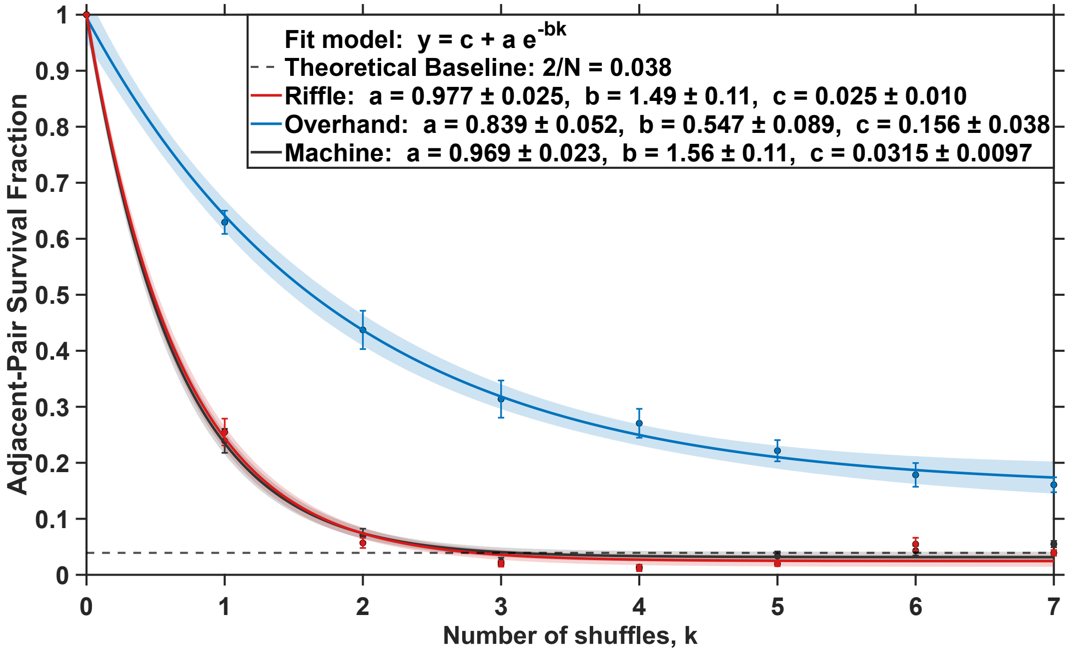


Figure 7: Adjacent-pair survival (APS) versus shuffle count for riffle, overhand, and machine shuffles. Mean APS values are plotted with standard error bars over ten independent trials. Solid curves represent exponential fits of the form $y = c + ae^{-bx}$, and shaded regions denote the 95 % confidence interval of the fitted model derived from all data points. The dashed line indicates the theoretical random-deck baseline of $2/N = 0.038$. Riffle and machine shuffles exhibited similar exponential decay rates of $b_{\text{riffle}} = 1.49 \pm 0.11$ and $b_{\text{machine}} = 1.56 \pm 0.11$, reaching the random baseline after roughly 4 shuffles. In contrast, the overhand shuffle decayed more slowly with $b_{\text{overhand}} = 0.547 \pm 0.089$ and remained approximately 4.18 times higher than the theoretical baseline after seven shuffles, indicating substantially slower mixing behavior.

These results confirm that both manual and mechanical riffle-style shuffles achieve near-random ordering within four to five shuffles which is faster than the theoretical “seven riffle” mixing threshold proposed by Bayer and Diaconis [2]. This apparent acceleration likely arises because the APS metric captures local pairwise mixing rather than full permutation entropy; local adjacency breaks down sooner than complete deck randomization. The similarity between the human riffle and the automated machine also suggests that the primary randomization mechanism, the interleaving of adjacent halves, is preserved regardless of operator variability, whereas the overhand method inherently maintains directional bias and weak interleaving between cards.

4.3 Normalized Total Variation Distance

Global randomness was evaluated using the Total Variation (TV) distance between the measured per-card positional distributions and the uniform random model. Because each method was performed across ten independent trials, the measured distributions represented a finite ensemble rather than an infinite sampling of all possible permutations. To correct for the finite number of trials $R = 10$, a normalization was applied such that complete randomization within those R observations corresponded to zero variation distance. The normalized TV was defined in Eq. (3) as

$$TV_{norm}(k) = \frac{TV(k) - TV_{floor}(R)}{TV_{max} - TV_{floor}(R)} \quad (3)$$

where Eq. (4) defines $TV_{floor}(R)$ as

$$TV_{floor}(R) = \frac{N - min(R,N)}{N} \quad (4)$$

and Eq. (5) defines TV_{max} as

$$TV_{max} = \frac{N-1}{N} \quad (5)$$

This scaling ensures that $\text{TV}_{norm} = 1$ represents a perfectly ordered deck, while $\text{TV}_{norm} = 0$ corresponds to complete randomization among the ten sampled trials. The resulting normalized TV curves for each shuffle type are shown in Figure 8, which compares the rate and extent of global randomization across methods. The figure highlights how rapidly the riffle and machine shuffles converge toward uniform randomness relative to the slower, plateauing overhand shuffle.

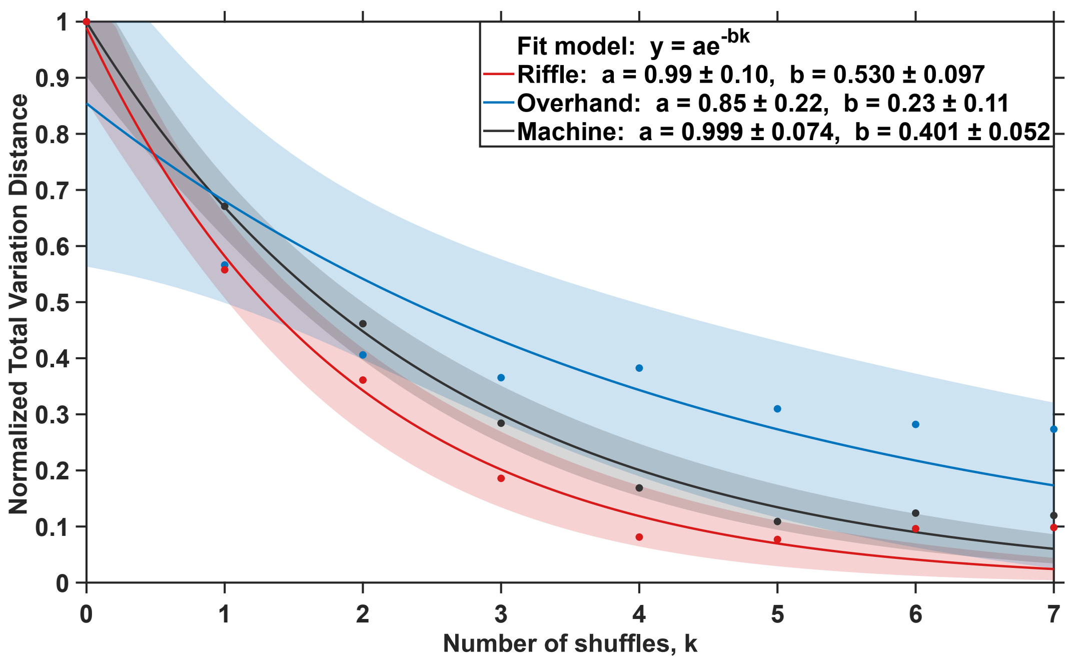


Figure 8: Normalized Total Variation (TV) distance versus shuffle count for riffle, overhand, and machine shuffles. Solid curves represent exponential fits of the form $y = ae^{-bk}$, and shaded regions denote the 95 % confidence intervals of the fitted models. Riffle and machine shuffles exhibited faster exponential decay with $b_{\text{riffle}} = 0.530 \pm 0.097$and $b_{\text{machine}} = 0.401 \pm 0.052$, while the overhand shuffle decayed more slowly with $b_{\text{overhand}} = 0.23 \pm 0.11$, indicating that even after seven iterations, significant residual structure persists in the deck's positional distribution.

These global-mixing results extend the local-order trends observed in the APS analysis (Section 4.2). Whereas APS captured the decay of adjacent-pair correlations after roughly four shuffles, the Total Variation (TV) metric measures convergence of the entire permutation space and thus continues decreasing until the deck approaches uniform randomness. The experimental riffle and machine curves reach their asymptotic minima near $k \approx 7$, in strong agreement with the Bayer and Diaconis prediction that total variation for an ideal riffle shuffle falls close to zero after about seven iterations [2]. The absence of a sharp cutoff in the experimental data arises from finite-sampling noise and natural trial-to-trial variability rather than any fundamental departure from theory.

Consistent with Pemantle's theoretical work, which estimated that an overhand shuffle would require on the order of a thousand repetitions to achieve statistical randomness [3], the present data show substantial residual order within the overhand system. At $k = 7$, the overhand shuffle's fit-predicted normalized TV value remains approximately 2.9 times higher than that of the machine shuffle and 7.2 times higher than that of the riffle, indicating that the deck retains substantial structure under overhand shuffling, whereas the other methods have already moved much closer to randomness.

Interestingly, the human riffle exhibits a steeper TV randomization rate ($\mathrm{b}_{\mathrm{riffle}} = 0.530 \pm 0.0097$) than the mechanical shuffle ($\mathrm{b}_{\mathrm{machine}} = 0.401 \pm 0.052$), suggesting that small variations in human shuffling can promote deeper card mixing. The mechanical shuffler, while highly repeatable, performs nearly identical interleaves each cycle and therefore converges at a slightly lower randomization rate.

Overall, the TV analysis demonstrates that both riffle-style methods achieve near-uniform global order within seven shuffles, validating classical theoretical predictions under real experimental conditions, whereas the overhand shuffle remains inefficient, diffusive, and directionally biased even after repeated mixing. Taken together with the APS findings, these results indicate that local order breaks down after roughly four shuffles, while full global randomness emerges only after about seven, a result fully consistent with the theoretical benchmark.

4.4 Limitations and Future Work

While the APS and TV analyses confirm that both human and mechanical riffle shuffles achieve near-uniform randomness within seven iterations, some limitations remain in the current experimental design. The use of ten independent trials per method provided a sufficient basis for curve-fitting and uncertainty estimation, but the normalized TV results (Fig. 8) showed finite-sampling noise that likely obscures the true asymptotic behavior near complete randomization. Expanding the number of trials to twenty or more would reduce this noise floor and sharpen the decay trends, allowing clearer identification of whether a sharp "cutoff" like that predicted by Bayer and Diaconis truly emerges in experimental data [2].

The fitted randomization rates quantify these effects numerically. Riffle shuffles showed $b_{\mathrm{riffle}} = 0.530 \pm 0.097$while the automatic machine exhibited $b_{\mathrm{machine}} = 0.401 \pm 0.052$, indicating that the human-performed riffle mixed roughly 30 % faster. This difference likely arises from subtle variations in human packet release that promote more interleaving, essentially, "imperfect" hands may create deeper mixing than the mechanical device's highly repeatable motion. In contrast, the overhand shuffle's slow decay constant $b_{\mathrm{overhand}} = 0.23 \pm 0.11$ confirms that its sequential packet transfer produces very limited diffusion; even after seven shuffles, its normalized TV remained ≈ 2.9 × higher than the machine and ≈ 7.2 × higher than the riffle.

These distinctions highlight how randomness depends not only on the shuffle type but also on the degree of human variability. Human riffles contain inherent bias with tendencies toward uneven packet sizes and release timing, yet those same irregularities can accelerate randomization compared with the predictable periodicity of a mechanical shuffler. In fact, a perfect riffle shuffle would not lead to randomness at all as the state could be predicted. Overhand shuffling, by contrast, is both biased and inefficient, preserving directional patterns and local structure that persist even after repeated mixing. Future studies could expand the trial set, but this project serves as a reliable baseline for physical shuffling systems.

5. Conclusions

A computer vision approach to measuring randomness in physical card shuffling was taken for this experiment. Using ArUco-tracked decks and automated analysis, the study quantified both local and global mixing behavior for riffle, overhand, and machine shuffles across 210 total trials. Adjacent-pair survival and normalized total-variation distance provided complementary insight: local order decayed after roughly four shuffles, while full-deck randomness emerged near seven, which is consistent with classical theory [2].

Riffle and mechanical shuffles showed comparable performance, with randomization rates of $b_{\text{riffle}} = 1.49 \pm 0.11$(APS) and 0.530 ± 0.097(TV) for the human riffle versus $b_{\text{machine}} = 1.56 \pm 0.11$(APS) and 0.401 ± 0.052(TV) for the machine. Overhand shuffling was significantly slower, $b_{\text{overhand}} = 0.547 \pm 0.089$(APS) and 0.23 ± 0.11(TV), validating theoretical predictions that it mixes far less efficiently than riffle-type methods [7].

In practical terms, these findings suggest that about four riffle shuffles are sufficient for everyday play, enough to eliminate visible order and achieve near-random hands, while a full seven ensures complete statistical mixing consistent with professional or casino-level fairness. The framework developed here demonstrates how computer vision and ArUco markers are a viable tracking metric for analyzing large quantities of physical data and provide insights into automating visually intensive analysis processes. With just a few quick shuffles, order collapses into chaos and the cards become fair to play with. This work also shows that accessible computer-vision tools can bridge physical experiments with mathematical models of randomness, and the same tracking framework could be applied to other physical mixing or ordering processes where manual data collection is impractical. In the end, the results show that randomness is not merely left to chance, it can be measured, shaped, and engineered with precision.

Acknowledgments

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